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Experiment – 9: AJAX

1) **Aim:** To study AJAX

2) **Problem Statement:**

Create a registration page having fields like Name, College, Username and Password (read password twice).

Validate the form by checking for

1. Username is not same as existing entries

2. Name field is not empty

3. Retyped password is matching with the earlier one. Prompt a message is

And also auto suggest college names.

Show the message "Successfully Registered" on the same page below the submit button, on Successful registration. Let all the updations on the page be Asynchronously loaded. Implement the same using XMLHttpRequest Object.

3) **Theory:**

1.How do Synchronous and Asynchronous Requests differ?

Feature	Synchronous Request	Asynchronous Request
Execution	Blocks the execution of code until request completes.	Does not block the code; continues execution.
Performance	Slower; user must wait until the request finishes.	Faster; allows other tasks to run simultaneously.
User Experience	Poorer, especially with slow network or large files.	Better, as the UI stays responsive.

Usage	Rarely used; not recommended in modern apps.	Commonly used in web applications.
Syntax in XMLHttpRequest	xhr.open("GET", url, false) synchronous	xhr.open("GET", url, true) asynchronous

2. Describe various properties and methods used in XMLHttpRequest Object

Methods

Method	Description
open(method, url, async)	Initializes a request. • method: GET, POST, etc. • url: Server URL • async: true (async) or false (sync)
send(body)	Sends the request to the server. Use null for GET, or pass data (string) for POST.
setRequestHeader(header, value)	Sets an HTTP request header (used after open, before send).
abort()	Cancels an ongoing request.
getResponseHeader(header)	Returns the value of a specific response header.
getAllResponseHeaders()	Returns all response headers as a string.

Properties

Property	Description
readyState	Holds the status of the request: 0 - UNSENT 1 - OPENED 2 - HEADERS_RECEIVED 3 - LOADING 4 - DONE
status	The HTTP status code of the response (e.g., 200, 404).

statusText	The HTTP status message (e.g., "OK", "Not Found").
responseText	The response data as a text string.
responseXML	The response data as an XML document (if response is XML).
Onreadystatechange	An event handler function that runs every time readyState changes.
responseType	Specifies the response type (e.g., "", "text", "json", "blob").
timeout	Sets the time in milliseconds for a request to wait before timing out.
withCredentials	Boolean that indicates whether cross-site requests should include credentials (cookies, headers).

4) Output:

Code-

Index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>AJAX Registration Experiment</title>
  <style>
    /* Reset & Base */
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body {
      display: flex;
      align-items: center;
      justify-content: center;
      min-height: 100vh;
      background: linear-gradient(135deg, #f5f7fa, #c3cfe2);
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    }

    /* Card Container */
    .card {
      background: #ffffff;

```

```
padding: 2rem;
border-radius: 1rem;
box-shadow: 0 8px 20px rgba(0, 0, 0, 0.1);
width: 100%;
max-width: 380px;
text-align: center;
}
```

```
.card h2 {
margin-bottom: 1.5rem;
color: #333;
font-size: 1.5rem;
}
```

```
/* Form Fields */
```

```
form {
display: flex;
flex-direction: column;
gap: 1rem;
}
```

```
input {
padding: 0.75rem 1rem;
border: 1px solid #ddd;
border-radius: 0.5rem;
font-size: 1rem;
transition: border-color 0.2s, box-shadow 0.2s;
}
```

```
input:focus {
border-color: #6c63ff;
box-shadow: 0 0 3px rgba(108, 99, 255, 0.2);
outline: none;
}
```

```
button {
padding: 0.75rem;
background: #6c63ff;
color: #fff;
border: none;
border-radius: 0.5rem;
font-size: 1rem;
cursor: pointer;
transition: background 0.2s, transform 0.1s;
}
```

```

button:hover {
  background: #5a54d1;
}

button:active {
  transform: scale(0.98);
}

/* Message Styles */
.message {
  margin-top: 1rem;
  font-weight: 600;
  font-size: 0.95rem;
}

.error {
  color: #e74c3c;
}

.success {
  color: #27ae60;
}
</style>
</head>
<body>

<div class="card">
  <h2>Register</h2>
  <form id="regForm">
    <input type="text" id="name" placeholder="Name" />
    <input type="text" id="college" placeholder="College" list="colleges" />
    <datalist id="colleges"></datalist>

    <input type="text" id="username" placeholder="Username" />
    <input type="password" id="password" placeholder="Password" />
    <input type="password" id="retype" placeholder="Retype Password" />

    <button type="submit">Register</button>
  </form>
  <div id="message" class="message"></div>
</div>

<script>
  // --- Data for simulation ---
  const existingUsernames = ["shree", "aaro", "arjun", "rishita"];
  const collegeList = [

```

```
"IIT Bombay", "IIT Delhi", "IIT Madras",  
"NIT Trichy", "IISc Bangalore", "BITS Pilani",  
"VIT Vellore", "SRM University", "Delhi University",  
"Anna University"  
];
```

```
// Populate datalist  
const dl = document.getElementById('colleges');  
collegeList.forEach(c => {  
  const opt = document.createElement('option');  
  opt.value = c;  
  dl.appendChild(opt);  
});
```

```
// Form submission handler  
document.getElementById('regForm').addEventListener('submit', function(e) {  
  e.preventDefault();  
  const name = document.getElementById('name').value.trim();  
  const college = document.getElementById('college').value.trim();  
  const username = document.getElementById('username').value.trim();  
  const password = document.getElementById('password').value;  
  const retype = document.getElementById('retype').value;  
  const msgDiv = document.getElementById('message');
```

```
// reset  
msgDiv.textContent = "";  
msgDiv.className = 'message';
```

```
// Create AJAX request (simulation)  
const xhr = new XMLHttpRequest();  
xhr.open('POST', '#', true); // '#' placeholder, no real endpoint  
xhr.setRequestHeader('Content-Type', 'application/x-www-form-urlencoded');
```

```
xhr.onreadystatechange = function() {  
  if (xhr.readyState === 4) {  
    // Simulate server processing delay  
    setTimeout(() => {  
      // 1. Name not empty  
      if (!name) {  
        msgDiv.textContent = 'Name field cannot be empty';  
        msgDiv.classList.add('error');  
        return;  
      }  
      // 2. Unique username  
      if (existingUsernames.includes(username)) {  
        msgDiv.textContent = 'Username already exists. Choose another';
```

```

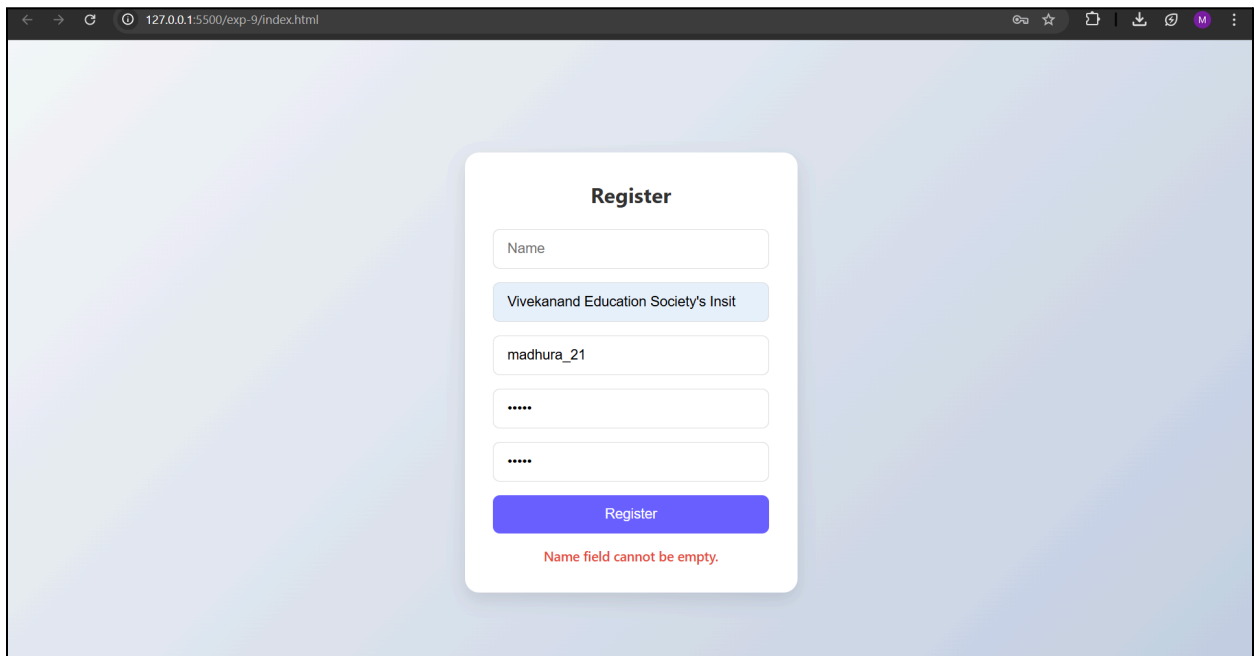
        msgDiv.classList.add('error');
        return;
    }
    // 3. Passwords match
    if (password !== retype) {
        msgDiv.textContent = 'Passwords do not match.';
        msgDiv.classList.add('error');
        return;
    }
    // All validations passed
    existingUsernames.push(username);
    msgDiv.textContent = 'Successfully Registered';
    msgDiv.classList.add('success');
}, 300);
}
};

// Send simulated data
const params =
`name=${encodeURIComponent(name)}&college=${encodeURIComponent(college)}&username
=${encodeURIComponent(username)}&password=${encodeURIComponent(password)}`;
    xhr.send(params);
});
</script>

</body>
</html>

```

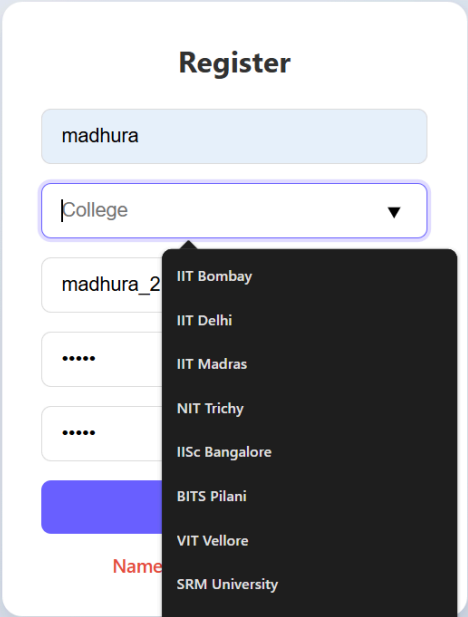
Name field validation-



The screenshot shows a web browser window with the address bar displaying '127.0.0.1:5500/exp-9/index.html'. The main content area features a 'Register' form. The form has the following fields and elements:

- Name:** A text input field that is currently empty.
- College:** A dropdown menu with 'Vivekanand Education Society's Insit' selected.
- Username:** A text input field containing 'madhura_21'.
- Password:** A text input field with masked characters '*****'.
- Retype Password:** A text input field with masked characters '*****'.
- Register:** A blue button.
- Error Message:** A red text message below the button stating 'Name field cannot be empty.'

Suggesting College entries-



The image shows a 'Register' form with the following fields: a text input for 'madhura', a dropdown menu for 'College', a text input for 'madhura_2', two password inputs (one with 5 dots, one with 4 dots), and a blue 'Register' button. A dropdown menu is open from the 'College' field, displaying a list of institutions: IIT Bombay, IIT Delhi, IIT Madras, NIT Trichy, IISc Bangalore, BITS Pilani, VIT Vellore, SRM University, Delhi University, Anna University, and Vivekanand Education Society's I... The text 'Name' is partially visible in red below the button.

Register

madhura

College ▼

madhura_2

.....

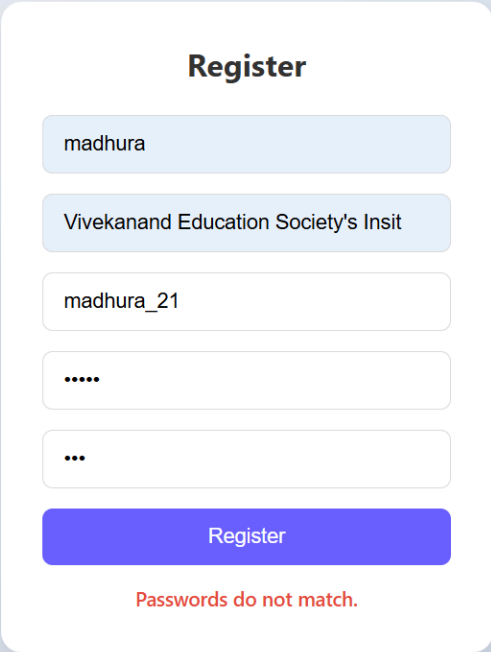
.....

Register

Name

- IIT Bombay
- IIT Delhi
- IIT Madras
- NIT Trichy
- IISc Bangalore
- BITS Pilani
- VIT Vellore
- SRM University
- Delhi University
- Anna University
- Vivekanand Education Society's I...

Checking Password-



The image shows the same 'Register' form as above, but with the following changes: the 'College' dropdown is now filled with 'Vivekanand Education Society's Insit', the second password input has 5 dots, and a red error message 'Passwords do not match.' is displayed below the 'Register' button.

Register

madhura

Vivekanand Education Society's Insit

madhura_21

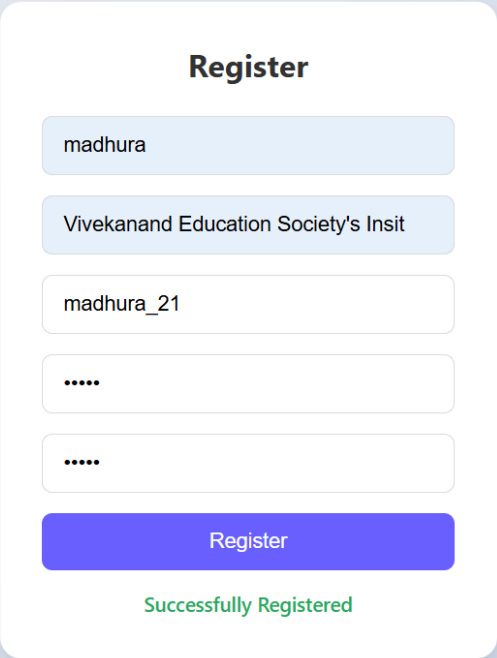
.....

...

Register

Passwords do not match.

Successfully Registered-



The image shows a 'Register' form on a light blue background. The form is a white card with rounded corners. It contains the following elements: a title 'Register' in bold black text; a light blue input field with the text 'madhura'; a light blue input field with the text 'Vivekanand Education Society's Insit'; a white input field with the text 'madhura_21'; two white input fields with masked text '.....'; a blue 'Register' button; and a green message 'Successfully Registered' at the bottom.

Register

madhura

Vivekanand Education Society's Insit

madhura_21

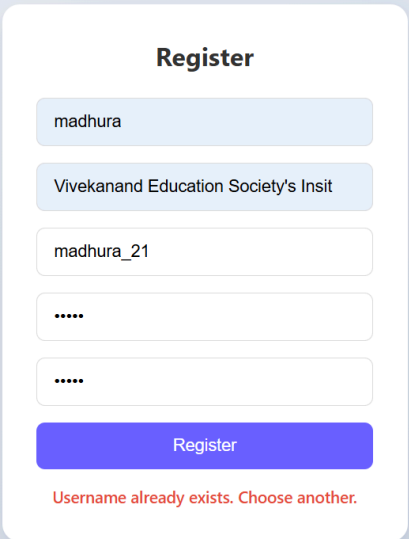
.....

.....

Register

Successfully Registered

Existing username not allowed-



The image shows the same 'Register' form as above, but with an error message. The 'Register' button is now disabled and greyed out. A red error message 'Username already exists. Choose another.' is displayed below the button.

Register

madhura

Vivekanand Education Society's Insit

madhura_21

.....

.....

Register

Username already exists. Choose another.